RESEARCH ARTICLE

Agricultural Production Planning under Correlated Price Risk: Exact Reference Computation and Surrogate-Guided Continuous Refinement

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ABSTRACT
Agricultural production planning couples discrete crop calendars with continuous land allocation under uncertain output prices. This study formulates a bi-objective mixed-integer planning model that maximizes expected income and minimizes covariance-based price risk while enforcing empirical crop eligibility, lot occupancy, maturation, and botanical-family recovery. Four publication-complete instances cover one or two farms, 2–8 lots, 5–12 crops, 52–104 planning weeks, and 0.6–6.0 ha. Exact ε-constraint runs in SCIP provide an incumbent reference, standard NSGA-II supplies a low-cost baseline, and two matheuristics apply either full or surrogate-guided continuous quadratic area refinement with SLSQP. Each stochastic method is evaluated over ten seeds using normalized hypervolume, IGD^+, additive epsilon, nondominated-set size, runtime, and refinement effort. Exact-run time rises from 51.25 s on the two-lot instance to 9.52 h on the largest completed instance, while most intermediate SCIP points on the larger instances terminate at the time limit. Surrogate gating reduces refinement calls from 12,000 to 3,000 per replication. On the 8-lot, 104-week instance, it reduces mean total runtime from 1,478.48 to 410.52 s relative to full refinement, but median hypervolume also declines from 0.745 to 0.718. Standard NSGA-II records the highest median hypervolume among the stochastic methods on three of four instances. The evidence therefore supports method selection as a trade-off among reference quality, computation, and planning scale, rather than a universal algorithm ranking.

KEYWORDS
agricultural production planning; multiobjective optimization; NSGA-II; matheuristics; surrogate-assisted optimization; price risk

1. Introduction

Crop planning commits scarce land and operational time before harvest prices become observable, while maturation delays and recovery requirements make many decisions costly to reverse. These commitments affect farm income, exposure to correlated market movements, and the productive continuity of heterogeneous lots, making planning consequential for both livelihoods and resource stewardship. Agri-food planning reviews accordingly emphasize that useful decision models must connect biological calendars, market conditions, and operational restrictions within a coherent temporal representation (Ahumada & Villalobos, 2009).

Production planning more generally has moved beyond deterministic cost minimization toward settings involving uncertain parameters, heterogeneous resources, inter-

dependent decisions, and several competing objectives. Exact multiobjective methods can provide status-bearing trade-offs, whereas evolutionary methods can explore extensive nonconvex decision spaces and return diverse alternatives within a single run (Deb et al., 2002; Mavrotas, 2009). However, growing combinations of products, resources, and periods make computational scalability inseparable from solution quality, particularly when mathematical subproblems are repeatedly embedded inside population-based search.

Agriculture intensifies these difficulties because planting and harvest dates are biologically linked, yield evidence varies geographically, crop families require recovery intervals, and prices co-move across markets. Existing models address uncertain maturation, demand, harvest, and yield (Tan & Comden, 2012), while portfolio-based farm planning connects diversification with financial and environmental exposure (Burbano-Figueroa et al., 2022; Rădulescu et al., 2014). Nevertheless, an implementable plan must integrate those economic trade-offs with weekly crop sequences and continuous land allocation rather than treating diversification as an aggregate acreage decision.

The resulting model contains a combinatorial scheduling layer and a structured continuous allocation layer, creating a natural opportunity for exact optimization, evolutionary search, and matheuristic decomposition. Production research shows that hybrid methods can preserve useful mathematical structure while extending practical scale (Battaïa & Dolgui, 2022; Hu et al., 2024), yet continuous refinement becomes burdensome when every evolutionary offspring receives an optimization call. Surrogate assistance can allocate expensive evaluations more selectively, although savings remain meaningful only when measured alongside changes in Pareto-front quality (Jin, 2011; Liu et al., 2024).

This study therefore addresses three questions that connect planning value, computational tractability, and the consequences of selectively allocating mathematical refinement across candidate agricultural schedules. RQ1 asks how rapidly exact ε-constraint reference computation becomes costly as farms, lots, crops, planning weeks, and feasible production events increase jointly. RQ2 asks how standard NSGA-II and two refinement-based matheuristics compare against status-aware SCIP incumbent archives using replicated Pareto-quality and computational-effort indicators. RQ3 asks whether surrogate-guided selection can reduce continuous-refinement effort while preserving enough frontier quality to offer a defensible alternative under constrained runtime budgets.

To answer these questions, the study formulates a weekly bi-objective event model maximizing expected income while minimizing covariance-based monetary price risk under empirical eligibility and agronomic constraints. The computational design compares SCIP ε-constraint search, standard NSGA-II, exhaustive continuous refinement, and surrogate-guided selective refinement using identical benchmark data and replicated stochastic seeds. Its contribution is a controlled, status-aware comparison that exposes certification, frontier quality, refinement calls, and runtime without claiming universal algorithmic superiority beyond the evaluated instances.

2. Related Research

2.1. Production planning and agricultural decisions under uncertainty

Contemporary production planning increasingly represents multiple performance criteria, resource heterogeneity, and uncertainty instead of relying exclusively on de-

terministic efficiency or cost objectives. Line-balancing reviews document expanding model variants, operational constraints, data requirements, and exact or heuristic solution families across increasingly heterogeneous industrial production environments (Battaïa & Dolgui, 2013; Boysen et al., 2022). Hybridization research further shows that increasingly structured production problems motivate combinations of mathematical programming, decomposition, and metaheuristics rather than one universally dominant solution paradigm (Battaïa & Dolgui, 2022).

Multiobjective production models require explicit treatment of trade-offs because improving one operational or economic criterion can degrade another criterion without yielding a single unequivocal optimum. The ε-constraint method produces interpretable alternatives by optimizing one objective while systematically bounding another, although its repeated mathematical programs can become computationally demanding (Mavrotas, 2009). Population-based alternatives such as NSGA-II instead approximate several nondominated solutions simultaneously through elitist sorting and diversity preservation, making replicated evaluation essential for stochastic comparisons (Deb et al., 2002).

Heterogeneous resources add more than scale because suitability, productivity, capacity, and timing determine which assignments are feasible and how strongly they contribute to each objective. Parallel-batch scheduling research illustrates how hybrid evolutionary mechanisms can coordinate assignment, batching, and delivery structures that resist straightforward exact solution at realistic sizes (Kucukkoc et al., 2024). This production-planning stream establishes the need for scalable multiobjective methods, but it does not determine how expensive mathematical improvement should be distributed among agricultural candidates.

Agricultural production planning joins tactical resource allocation with biological timing, uncertain market outcomes, spatially variable productivity, and operational decisions extending from planting through harvest and distribution. The foundational agri-food review by Ahumada & Villalobos (2009) identifies planning models across these linked decisions, while Tan & Comden (2012) explicitly represent demand, maturation, harvest, and yield risk for annual crops. These characteristics distinguish agricultural planning from generic allocation because feasible decisions depend on biological calendars that constrain when occupied land becomes available again.

Portfolio formulations provide a complementary economic perspective because crop returns are jointly distributed, meaning diversification depends on covariance rather than independent crop-specific penalties. Rădulescu et al. (2014) integrate expected return with financial and environmental risks, whereas Burbano-Figueroa et al. (2022) apply decision analysis and portfolio theory to diversification under horticultural uncertainty. Those studies motivate covariance-aware risk measurement, although their aggregate decision structures do not reproduce weekly lot occupancy and crop-family recovery requirements.

Recent agricultural multiobjective studies increasingly combine sustainability, distribution logistics, interactive decision support, and evolutionary search while retaining consequential application-specific agronomic constraints. Vergara Rodríguez et al. (2025) optimize cultivation and distribution decisions for Colombian yam systems, whereas Huang et al. (2025) incorporate fuzzy information and expert rules into an enhanced NSGA-II crop-planning framework. Kropp et al. (2026) add interactive decision making to evolutionary agricultural optimization and report competitive solution support with fewer evaluations, demonstrating growing interest in computationally economical decision interaction.

The present setting extends this stream by linking empirical municipality–crop eligi-

bility, event-level weekly calendars, continuous cultivated areas, and correlated harvest-price exposure within one reproducible production model. Its risk term remains deliberately bounded because it represents covariance-based price exposure rather than weather, yield, behavioural, or complete supply-chain uncertainty.

2.2. Multiobjective and matheuristic solution paradigms

Matheuristics exploit complementary strengths by assigning combinatorial exploration to a heuristic while mathematical programming resolves a structured subproblem or improves selected candidate solutions. In multiobjective cellular manufacturing, Hu et al. (2024) embed mathematical optimization within evolutionary scheduling, showing how exact structure can support rather than replace population-based search. This decomposition is especially relevant when fixing discrete identities and times converts the remaining allocation problem into a more tractable continuous program.

Production applications demonstrate several ways to connect mathematical and heuristic components, including formulation-driven neighborhoods, decomposition, route improvement, and coordinated scheduling decisions. Li et al. (2022) combine an efficient formulation with a matheuristic for unrelated parallel machines, while Qin et al. (2024) address capacitated multi-level lot sizing with substitution and backorders. Related designs improve production routing (Vadseth et al., 2023) and integrate production with distribution scheduling (Vié et al., 2025), confirming that hybrid structure can broaden computational reach across planning levels.

Industrial evidence also indicates that practical matheuristics must be judged through computational reliability and decision quality under realistic planning horizons rather than through algorithm labels alone. Seyfi & Yanıkoğlu (2026) integrate lot sizing, machine scheduling, shifts, and workforce planning for washing-machine production, obtaining scalable feasible solutions when longer horizons challenge direct optimization. These studies motivate conditional mathematical refinement, but they rarely isolate whether refining every evolutionary candidate is preferable to allocating refinement through learned utility.

2.3. Surrogate-guided optimization and research gap

Surrogate-assisted evolutionary computation replaces or prioritizes expensive evaluations using models trained on previously evaluated candidates, thereby treating computation as a resource-allocation decision. Early synthesis emphasizes that imperfect surrogates require explicit model management and continued contact with true evaluations to avoid misleading the search (Jin, 2011). Multiobjective surveys similarly identify accuracy, dimensionality, infill selection, and limited evaluation budgets as central design challenges for computationally expensive optimization (Chugh et al., 2019).

More recent reviews broaden this foundation by classifying surrogate models, algorithmic roles, applications, and unresolved questions across continuous and combinatorial expensive optimization. He et al. (2023) review surrogate-assisted evolutionary algorithms across several problem classes and applications, while Liu et al. (2024) show that expensive combinatorial settings remain substantially less developed than continuous counterparts. A recent differential-evolution review likewise stresses that real-world value depends on whether surrogate overhead remains small relative to the evaluations being rationed (Yu et al., 2025).

Current algorithmic developments improve relationship learning and local-region management for expensive multiobjective problems, but their customary goal remains prediction or replacement of expensive objective evaluations. Pierce et al. (2025) learn informative comparison relationships among candidate solutions, whereas Wang et al. (2026) adapt local-region search for challenging high-dimensional expensive multiobjective optimization problems within demanding practical settings. The present study instead predicts realized optimization gain and uses that estimate only to decide which candidates receive a true continuous-refinement call.

Across these four streams, no reviewed study jointly provides weekly agricultural scheduling, covariance-based monetary risk, exact status-aware references, exhaustive continuous refinement, and surrogate-gated refinement within one controlled experiment. This gap supports a focused contribution concerning the allocation of optimization effort, while avoiding broader claims that the underlying algorithms or agricultural constraints are individually unprecedented.

3. Decision Setting and Data

3.1. Planning setting and productivity evidence

The decision setting represents Colombian farms divided into heterogeneous lots. A production event assigns a crop, sowing week, harvest week, and cultivated area to one lot. The source productivity table contains 17,059 municipality–crop–planting-month observations, 17 municipalities, and 40 recorded agricultural products. These counts describe the source table; the completed benchmarks use supported subsets of 5–12 crops rather than all 40 products.

Yield estimation combines municipality, crop, and planting-period evidence. Crop eligibility requires at least three supporting observations for a lot's municipality–crop pair; if no crop passes that threshold for a generated lot, the crop with greatest support is retained to avoid a structurally unusable lot. Feasible event variables are created only when the sowing calendar permits the event, harvest occurs within the horizon, the empirical eligibility rule is satisfied, and expected yield is positive.

3.2. Prices and uncertainty

Weekly prices were constructed for Bucaramanga's Centroabastos market from DANE SIPSA wholesale workbooks for 2018–2026. The input provenance records the downloaded files and hashes. The experiment uses a 104-week price forecast generated from the processed weekly series; shorter instances use its first 52 weeks. A Ledoit–Wolf covariance estimate of crop log returns supplies a positive-semidefinite dependence matrix, with numerical symmetrization and a diagonal adjustment only if required for convexity.

The optimization is performed in Colombian pesos (COP). Publication tables and figures apply a frozen reporting rate of 3,306.86 COP/USD dated 28 September 2026. If p_{kt} is COP/kg, then $p_{kt}^{\mathrm{USD}} = p_{kt}/3306.86$. Likewise, income in million COP is converted to thousand USD by multiplication by 1000/3306.86, and risk based on million-COP exposure is converted by $(1000/3306.86)^2$. Positive constant rescaling leaves Pareto dominance unchanged; the quadratic risk unit changes by the square of the monetary conversion factor.

No `demand_weekly.csv` file was present in the reproducible computational snapshot.

Consequently, the reported experiment has no market-throughput cap and makes no use of SIPSA-A demand.

3.3. Benchmark construction

The benchmark ladder changes farms, lots, area, crop count, and horizon under fixed instance seeds. Table 1 reports the publication-complete portion of this ladder and preserves the configured progression from small single-farm cases toward larger heterogeneous planning settings.

Only instances with all expected files were admitted: one exact run and ten seeds (2026–2035) for each stochastic method. The snapshot scanner reads each `metadata.json`, requires `status=ok`, and verifies the expected front before inclusion. It does not rely on a final run index. At the analysed snapshot, 4 instances meet this rule; the largest contains 8 lots.

4. Mathematical Formulation

The decision problem selects agronomically feasible crop events and their areas. Expected income is maximized while correlated price risk is minimized. The formulation is expressed in the internal monetary units used by optimization; the ex-post USD transformation above is used for manuscript reporting.

4.1. Sets, parameters, and variables

Let $\mathcal{F}$ be the farms, $\mathcal{L}$ the lots, and $\mathcal{L}_f \subseteq \mathcal{L}$ the lots belonging to farm f. Let $\mathcal{K}$ denote crops, $\mathcal{T}$ weekly planning periods, and $\mathcal{T}_k \subseteq \mathcal{T}$ feasible harvest periods for crop k. For an optional market extension, $\mathcal{G}$ denotes sales groups. The sets $\mathcal{C}_{lw}$ and $\mathcal{C}^q_{lw}$ contain, respectively, crop–harvest events that occupy lot l during week w and events from botanical family q whose occupation or recovery intervals conflict in that week.

Parameter A_l is lot area (m^2); n_k is crop maturation duration (weeks); $s(k,t) = t - n_k$ is the sowing week implied by harvest t; r_{lks} is expected yield (kg/m^2); and p_{kt} is expected price (COP/kg). The indicator b_{lk} equals one when empirical evidence makes crop k eligible for lot l. Matrix $\Sigma \succeq 0$ is the covariance matrix of crop price log returns, $q(k)$ is botanical family, and $g(k)$ is sales group. The optional parameter D_{gt} would cap group-g throughput in week t, but is absent from the reported base experiment.

For every feasible event (l,k,t), define

$$y_{lkt} = \begin{cases} 1, & \text{if crop } k \text{ is activated on lot } l \text{ for harvest in week } t, \\ 0, & \text{otherwise,} \end{cases} \tag{1}$$

Table 1. Publication-complete benchmark instances.

Instance	Farms	Lots	Area (ha)	Crops	Weeks
I01_micro	1	2	0.6	5	52
I02_small	1	4	1.8	8	52
I03_diverse	1	8	4.5	12	104
I04_two_farms	2	8	6.0	8	52

and let $z_{lkt} \geq 0$ be cultivated area (m^2). Aggregate harvested quantity is $x_{kt} \geq 0$ (kg). Monetary exposure is $e_{kt} \geq 0$, measured in million COP, and $\rho \geq 0$ is the epigraph variable used for quadratic risk in the exact formulation.

4.2. Bi-objective production-planning model

Production and exposure are aggregated through

$$x_{kt} = \sum_{l \in \mathcal{L}} r_{lk,s(k,t)} z_{lkt}, \qquad k \in \mathcal{K},\ t \in \mathcal{T}_k, \tag{2}$$

$$e_{kt} = \frac{p_{kt} x_{kt}}{10^6}, \qquad k \in \mathcal{K},\ t \in \mathcal{T}. \tag{3}$$

The two objectives are

$$\max \quad F_1 = \sum_{k \in \mathcal{K}} \sum_{t \in \mathcal{T}} e_{kt}, \tag{4}$$

$$\min \quad F_2 = \sum_{t \in \mathcal{T}} \mathbf{e}_t^{\mathsf{T}} \Sigma \mathbf{e}_t, \tag{5}$$

where F_1 is expected income in million COP and F_2 is correlated price risk in (million COP)^2. The latter is a monetary mean–variance measure, not a production cost.

Because neither objective rewards continuous land occupation, reduced productive intensity can become an endogenous component of the income–risk compromise rather than an algorithmic failure. Indeed, the feasible zero-production schedule has zero monetary price exposure, while intermediate Pareto solutions can lower exposure through smaller cultivated areas, fewer events, or longer inactive intervals. Land use is therefore audited as an outcome rather than introduced retrospectively as a third objective or binding constraint.

For a decoded event e with cultivated area a_e, sowing week s_e, and harvest week h_e, area-adjusted space–time utilization is calculated as

$$U = \frac{\sum_e a_e (h_e - s_e)}{|\mathcal{T}| \sum_{l \in \mathcal{L}} A_l}. \tag{6}$$

This definition records partial-area cultivation directly, whereas lot-level temporal occupancy $U_l^{\text{time}} = |\cup_{e \in \mathcal{E}_l} [s_e, h_e)|/|\mathcal{T}|$ records the proportion of planning weeks containing any productive event on lot l.

Activation uses the physical lot area rather than an arbitrary large constant:

$$0 \leq z_{lkt} \leq A_l y_{lkt}, \qquad l \in \mathcal{L},\ k \in \mathcal{K},\ t \in \mathcal{T}_k. \tag{7}$$

Weekly conflict cliques enforce exclusive lot occupation,

$$\sum_{(k,t) \in \mathcal{C}_{lw}} y_{lkt} \leq 1, \qquad l \in \mathcal{L},\ w \in \mathcal{T}, \tag{8}$$

and family-specific cliques extend this restriction through the configured recovery in-

terval,

$$\sum_{(k,t)\in\mathcal{C}^q_{lw}} y_{lkt} \le 1, \qquad l \in \mathcal{L},\ w \in \mathcal{T},\ q. \tag{9}$$

Variables exist only if $b_{lk} = 1$, $s(k,t)$ is an allowed sowing week, t lies within the horizon, and $r_{lk,s(k,t)} > 0$. This event generation enforces empirical eligibility and temporal feasibility without adding redundant zero variables.
Optional market-throughput extension. If independent market-capacity evidence is available, the implementation can add

$$\sum_{k:g(k)=g} x_{kt} \le D_{gt}, \qquad g \in \mathcal{G},\ t \in \mathcal{T}. \tag{10}$$

Constraint (10) was inactive in every result reported here.

4.3. Epsilon-constraint reference formulation

The reference procedure first solves

$$I^{\max} = \max\{F_1 : (2) - -(9)\}. \tag{11}$$

For each configured $\eta \in \{0, 0.1, \ldots, 1\}$, SCIP then solves

$$\begin{aligned} \min \quad & \rho, \\ \text{s.t.} \quad & \rho \ge F_2, \\ & F_1 \ge \eta I^{\max}, \qquad\qquad (2) - -(9). \end{aligned} \tag{12}$$

When a feasible incumbent is found, an optional strong-Pareto tie break maximizes F_1 subject to the attained risk plus a numerical tolerance. Solver status and relative gap are retained for every point, so a feasible time-limited incumbent is not relabelled as certified optimal.

5. Solution Methods

5.1. Chromosome and common evolutionary backbone

Figure 1 summarizes how every evolutionary candidate combines discrete crop identities and waiting decisions with continuous cultivated-area fractions across aligned lot–slot matrices. The deterministic decoder converts those genes into dated production events while enforcing eligibility, temporal availability, botanical-family recovery, planning-horizon, and positive-output conditions before evaluating economic objectives. Idle weeks are not interpreted uniformly as inefficiency because sowing calendars, maturation, setup, same-family recovery, unused slots, and horizon truncation can all create structurally feasible gaps. Other gaps arise endogenously from waiting genes, which permit zero to four additional weeks before the next admissible sowing date and thereby alter harvest prices and correlated monetary exposure. Consequently, the

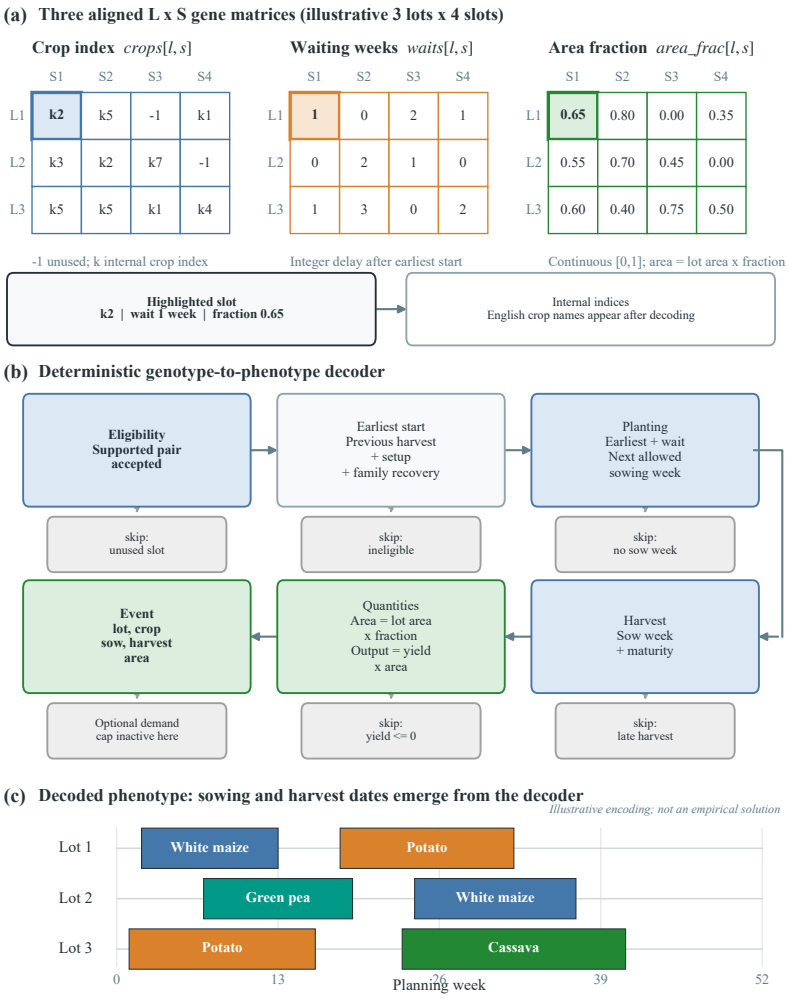


Figure 1. Chromosome semantics and deterministic decoding used by every evolutionary method. Aligned crop, waiting-week, and area-fraction matrices encode each candidate, while the decoder derives feasible sowing dates, harvest dates, cultivated areas, production quantities, and event records. The displayed schedule is illustrative rather than an empirical optimization result.

model distinguishes agronomically induced inactivity from economically chosen timing and production intensity without requiring continuous occupation of every lot.

The baseline encodes crop sequences and waiting decisions by lot. Decoding produces feasible sow–harvest events and area values, after which true income and risk are evaluated. Binary tournament selection, crossover probability 0.90, mutation, nondominated sorting, and crowding-based environmental selection operate with population 80 for 150 generations. Ten fixed seeds permit replication without changing instance data.

5.2. Continuous refinement variants

Baseline NSGA-II. Standard NSGA-II retains the decoded cultivated areas and objective values without invoking continuous mathematical refinement, thereby providing the common low-cost evolutionary reference for both hybrid variants.
Full refinement.

Conditional on a decoded discrete schedule, area allocation has a linear income expression, a positive-semidefinite quadratic risk expression, physical lot constraints, and fixed event timing. The reported NSGA-II+refinement experiment sends every offspring to this continuous subproblem. Although its mathematical structure is a convex quadratic refinement, the computational backend in this experiment is SLSQP, not SCIP.

5.3. Surrogate-guided refinement

H-NSGA-II uses the same evolutionary operators and SLSQP refinement, but limits refinement to 25% of offspring. During five warm-up generations, candidates are sampled for refinement. After at least 30 realized gains are available, an Extra Trees regressor ranks candidates; 20% of each refinement budget remains random exploration. All selection, nondominated sorting, and reported objectives use actual decoded or SLSQP-refined values. Surrogate predictions only control access to refinement.

Figure 2 isolates the principal algorithmic contrast by holding evolutionary generation and Pareto survival constant while changing only how conditional area refinement is allocated. The surrogate estimates realized refinement gain for candidate prioritization, whereas every selection decision and archived objective value remains grounded in decoded or SLSQP-refined evaluations.

6. Computational Experiment

Figure 3 connects empirical inputs, configured benchmark construction, four computational methods, controlled stochastic comparisons, evaluation indicators, and the publication-completeness filter applied to the reported snapshot. This workflow distinguishes the eight configured instances from the four publication-complete instances, thereby preventing partial larger-instance outputs from entering inferential comparisons.

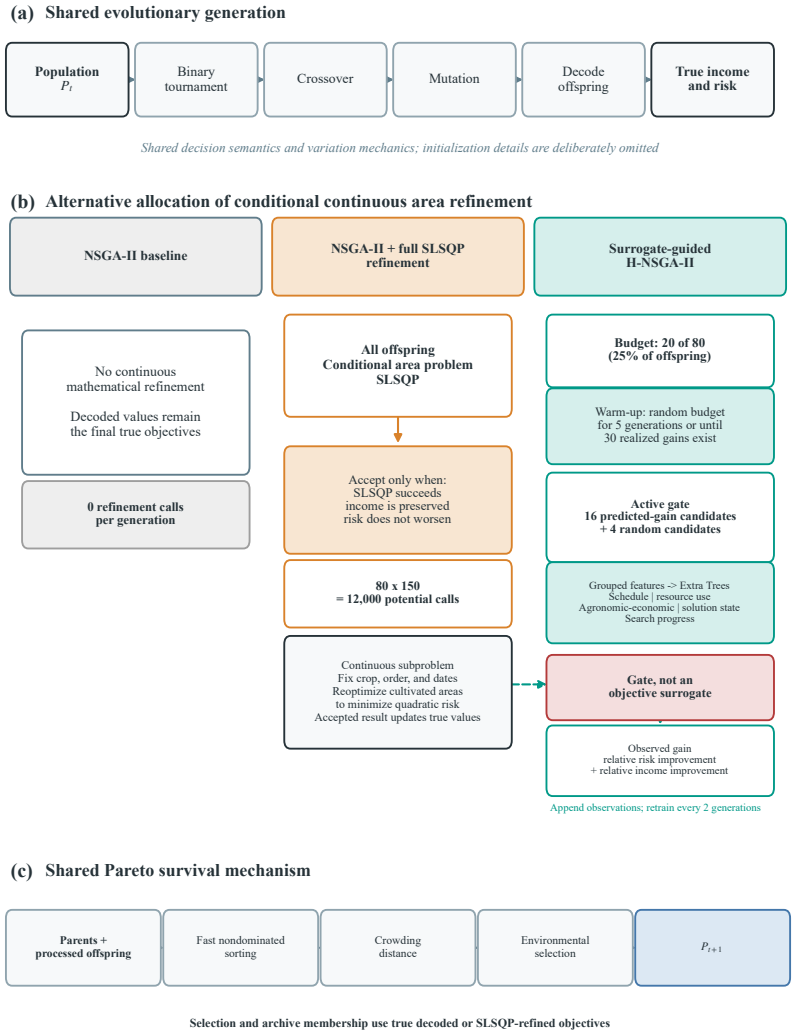


Figure 2. Shared NSGA-II backbone and alternative allocation of conditional continuous area refinement. Baseline NSGA-II performs no mathematical refinement, full refinement sends every offspring to SLSQP, and surrogate-guided H-NSGA-II allocates a 25% budget using predicted realized gain plus random exploration. All variants retain true objective values for Pareto survival.

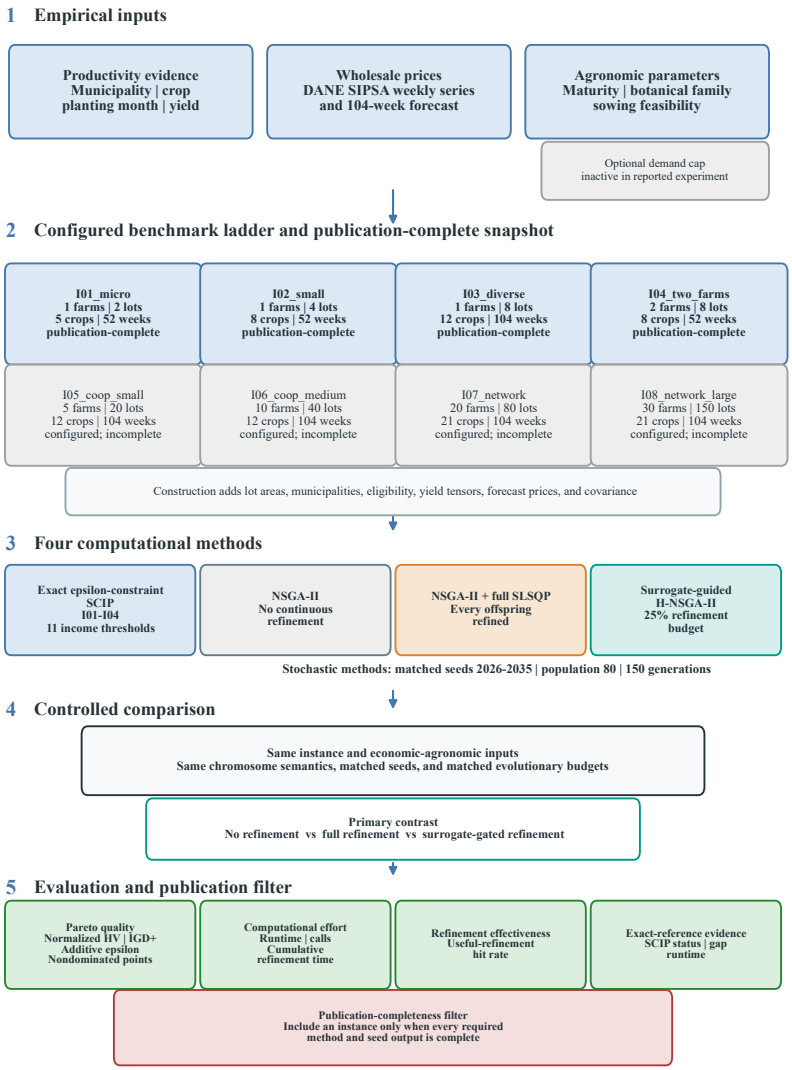


Figure 3. End-to-end computational and experimental workflow for the publication snapshot. Empirical productivity, wholesale-price, and agronomic inputs construct eight configured benchmarks; four complete instances enter exact and stochastic comparisons under matched controls, followed by Pareto-quality, computational-effort, refinement-effectiveness, and exact-reference evaluation.

6.1. Experimental design

All results come from `results/publication_slsqp_test`. Complete instances contain one exact run and 30 stochastic runs: ten each for NSGA-II, full refinement, and H-NSGA-II. Incomplete instances are excluded as units; a finished seed cannot enter tables unless every required seed and method for that instance is present. Optimization outputs remain unchanged, and manuscript currency conversion is reporting-only.

SCIP solves the mixed-integer quadratically constrained formulation at 11 income thresholds for I01–I04, using four threads and a target relative gap of 10^{-4}. Instance-specific limits are 300, 600, 1,200, and 1,800 s per solve, while the optional strong-Pareto tie break can require a second solve and therefore raise total runtime beyond 11 times the nominal point limit.

6.2. Performance indicators

For I01–I04, the nondominated SCIP incumbent archive defines the normalization and reference set. This label is deliberate because time-limited points may have nonzero or uninformative gaps. Hypervolume uses reference point $(1.1, 1.1)$ after objective normalization; higher values indicate greater dominated normalized area. Lower IGD^{+} and additive epsilon are preferred.

Nondominated point count, total runtime, refinement calls, cumulative refinement time, and useful-refinement hit rate complement these quality measures. Medians summarize stochastic quality and runtime; mean runtime is used only for explicitly stated runtime ratios.

7. Results

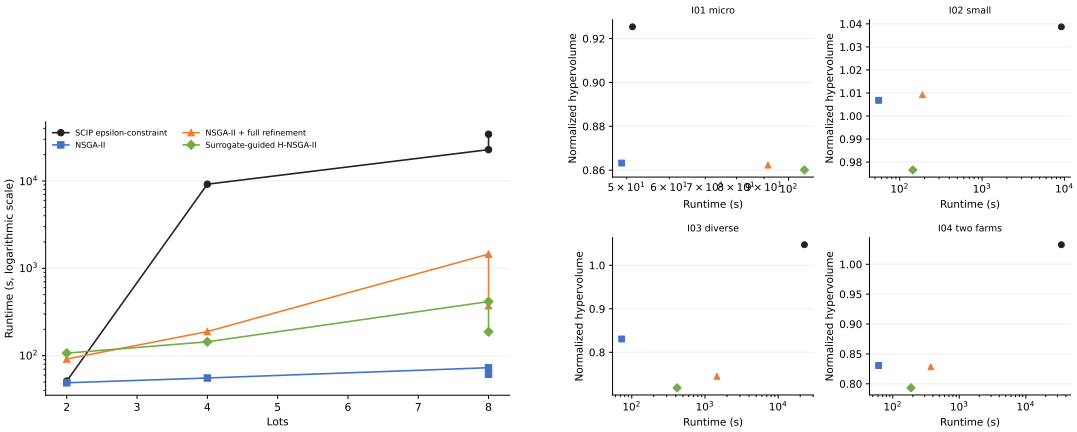
7.1. Exact computation and scalability

Total exact-run time increases from 51.25 s for I01 to 9,185.10 s (2.55 h) for I02, 22,824.80 s (6.34 h) for I03, and 34,274.58 s (9.52 h) for I04. Status evidence changes with scale. Ten of 11 I01 points are optimal and one terminates at the target gap. I02 has four optimal and seven time-limited points. I03 and I04 each have one optimal income-endpoint and ten time-limited points. Thus, SCIP remains valuable as an incumbent reference, but the experiment does not establish full-front optimality beyond the smallest instance.

NSGA-II median runtime stays between 48.93 and 72.84 s across completed instances, whereas exact runtime grows by nearly three orders of magnitude. Full refinement raises median runtime to 1,454.71 s on I03 and 372.73 s on I04. H-NSGA-II reduces those medians to 416.35 and 187.86 s, respectively. Using means, full refinement is 3.60 times slower than H-NSGA-II on I03 and 1.97 times slower on I04.

Table 2. SCIP epsilon-constraint run and point status.

Instance	Runtime (s)	Points	Optimal	Gap limit	Time limit	Max gap
I01_micro	51.25	11	10	1	0	9.85e-06
I02_small	9185.10	11	4	0	7	1.773
I03_diverse	2.28e+04	11	1	0	10	2.376
I04_two_farms	3.43e+04	11	1	0	10	1.394



(a) Runtime scaling

(b) Quality versus time

Figure 4. Computational effort and normalized hypervolume across complete instances.

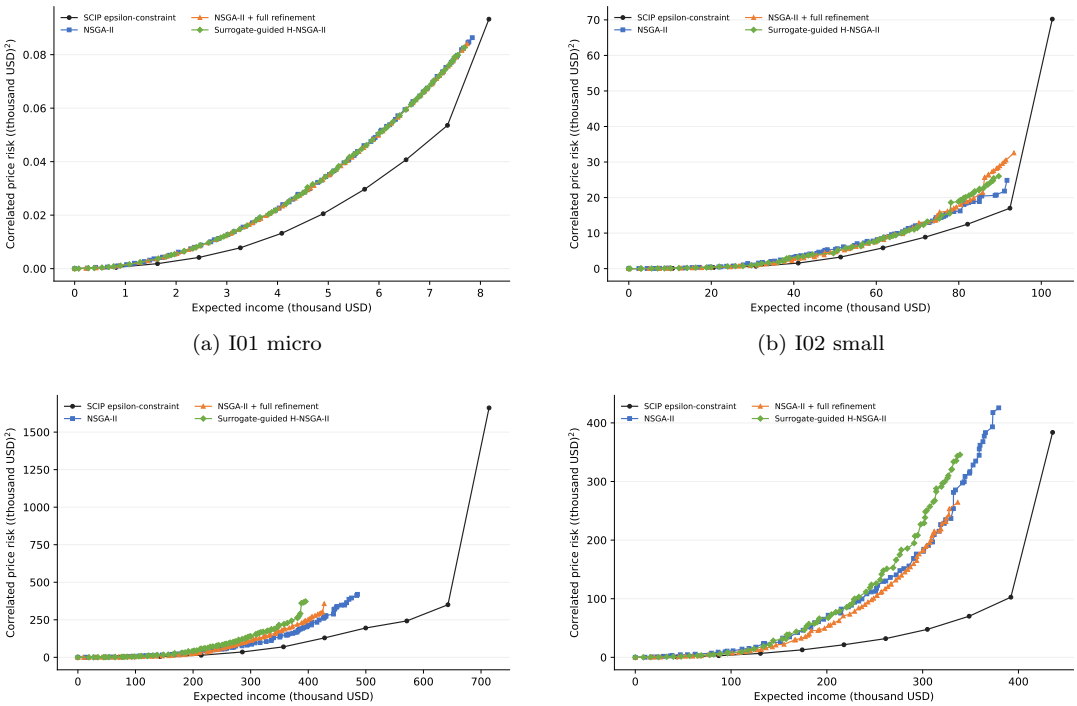
7.2. Pareto quality

Table 3 reports seed medians. Standard NSGA-II has the highest stochastic median hypervolume on I01 (0.863), I03 (0.831), and I04 (0.831). Full refinement is marginally highest on I02 (1.009 versus 1.007 for NSGA-II), with slightly lower median IGD^+ (0.030 versus 0.032). These differences are descriptive, not evidence of a universal ordering. On I03, full and selective refinement attain lower median hypervolume than NSGA-II, while on I04 full refinement is close to NSGA-II and H-NSGA-II is lower. Conditional area optimization therefore does not translate automatically into a better final archive under fixed evolutionary settings.

Figure 5 shows representative seeds chosen nearest each method's median hypervolume. The SCIP curves are incumbent archives; points with time-limit status must not be read as proofs of global efficiency.

Table 3. Median Pareto quality and runtime; stochastic methods use ten seeds.

Instance	Method	HV	IGD^+	ϵ^+	ND points	Runtime (s)
I01_micro	SCIP	0.925	0.000	0.000	11.0	51.25
I01_micro	NSGA-II	0.863	0.062	0.089	78.0	48.93
I01_micro	NSGA-II + full	0.862	0.063	0.090	78.0	91.51
I01_micro	H-NSGA-II	0.860	0.063	0.089	78.0	106.98
I02_small	SCIP	1.039	0.000	0.000	11.0	9185.10
I02_small	NSGA-II	1.007	0.032	0.109	79.0	55.51
I02_small	NSGA-II + full	1.009	0.030	0.109	78.5	188.85
I02_small	H-NSGA-II	0.977	0.040	0.138	78.5	144.15
I03_diverse	SCIP	1.047	0.000	0.000	11.0	2.28e+04
I03_diverse	NSGA-II	0.831	0.081	0.306	79.5	72.84
I03_diverse	NSGA-II + full	0.745	0.104	0.400	80.0	1454.71
I03_diverse	H-NSGA-II	0.718	0.121	0.421	79.5	416.35
I04_two_farms	SCIP	1.033	0.000	0.000	11.0	3.43e+04
I04_two_farms	NSGA-II	0.831	0.116	0.210	79.0	61.28
I04_two_farms	NSGA-II + full	0.829	0.112	0.222	79.0	372.73
I04_two_farms	H-NSGA-II	0.793	0.127	0.245	79.0	187.86



(a) I01 micro

(b) I02 small

(c) I03 diverse

(d) I04 two farms

Figure 5. Income–risk fronts in publication USD units. Stochastic lines show a median-hypervolume representative seed.

7.3. Refinement efficiency and decision interpretation

Full refinement makes 12,000 SLSQP calls per replication; surrogate gating makes 3,000, a 75% reduction. On I03, median cumulative refinement time falls from 1,346.04 to 284.46 s and useful-refinement hit rate rises from 96.0% to 98.8%. On I04, refinement time falls from 289.67 to 71.54 s and hit rate rises from 94.8% to 98.9%. The higher hit rate shows concentrated allocation, but quality must be read jointly: median hypervolume falls from 0.745 to 0.718 on I03 and from 0.829 to 0.793 on I04. Surrogate gating is therefore a computational economy with a measured quality cost in these instances.

A decision-level example was selected from the median-hypervolume H-NSGA-II seed for I02. Within that front, the compromise minimizes Euclidean distance to the normalized ideal of maximum income and minimum risk. The selected point yields expected income of 59.46 thousand USD, risk of 7.52 (thousand USD)^2, and area-

Table 4. Full versus surrogate-guided SLSQP refinement.

Instance	Method	Calls	Refinement time (s)	Useful (%)	HV
I01_micro	NSGA-II + full	1.20e+04	29.36	62.3	0.862
I01_micro	H-NSGA-II	3000	9.69	86.2	0.860
I02_small	NSGA-II + full	1.20e+04	114.72	90.4	1.009
I02_small	H-NSGA-II	3000	33.92	98.0	0.977
I03_diverse	NSGA-II + full	1.20e+04	1346.04	96.0	0.745
I03_diverse	H-NSGA-II	3000	284.46	98.8	0.718
I04_two_farms	NSGA-II + full	1.20e+04	289.67	94.8	0.829
I04_two_farms	H-NSGA-II	3000	71.54	98.9	0.793

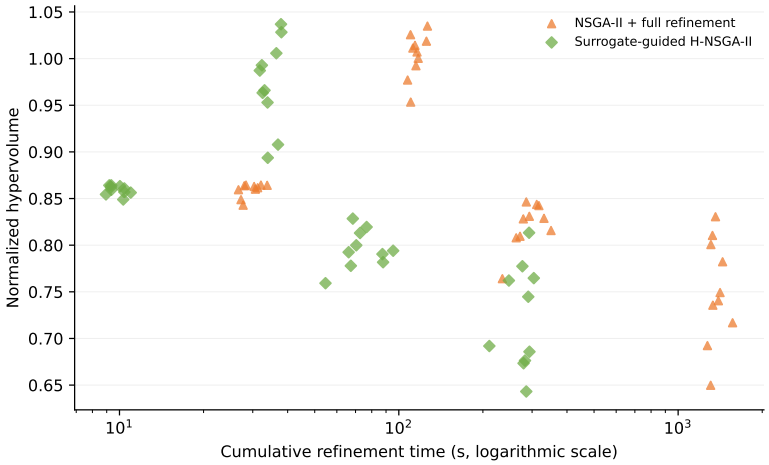


Figure 6. Normalized hypervolume against cumulative SLSQP refinement time for full and surrogate-guided refinement.

adjusted space–time utilization of 51.2%. Figure 7 translates that point into 11 production events over four lots, with mean temporal occupancy of 69.2% and mean idle time of 16.0 weeks per lot. The difference between area-adjusted utilization and temporal occupancy reflects partial-area events, whose allocated areas range from approximately 0.03 to 0.70 ha. These diagnostics show that white space in the Gantt combines agronomic feasibility, discretionary timing, and reduced productive intensity rather than representing a single form of scheduling inefficiency.

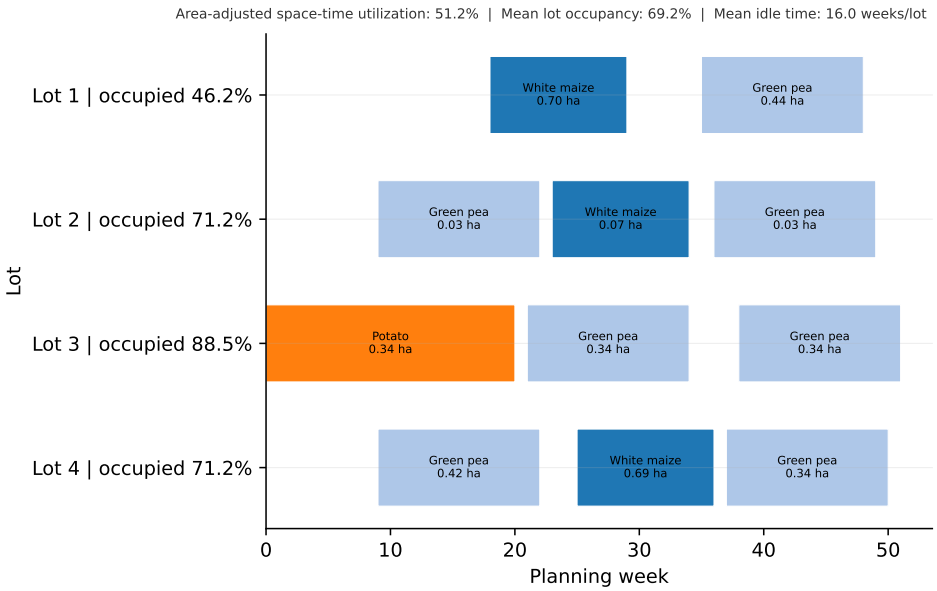


Figure 7. Representative compromise schedule for I02 small. Bars span sowing through the week preceding harvest, internal labels report crop and cultivated area, and lot labels report temporal occupancy. The upper annotation distinguishes area-adjusted space–time utilization from mean lot occupancy and reports mean inactive weeks per lot.

8. Discussion

The findings address a narrower gap than introducing an entirely new optimization family, because their main value lies in connecting agricultural scheduling with controlled refinement allocation. Relative to agricultural multiobjective models and production matheuristics, the experiment makes the computational consequence of exhaustive versus surrogate-gated continuous improvement directly observable under identical data, operators, and seeds.

The experiment separates three computational roles. SCIP supplies economically interpretable reference points and status information, but its rapid runtime growth and the prevalence of time-limited incumbents restrict certification beyond I01. Standard NSGA-II supplies diverse fronts at low and comparatively stable cost. Conditional continuous refinement exploits convex area structure, yet more local optimization does not guarantee a better global archive because refinement changes the search trajectory and consumes time without expanding the discrete schedule space.

Surrogate-guided refinement succeeds at its immediate task: it directs one quarter as many SLSQP calls toward candidates with a higher useful-refinement rate. The larger completed instances show the practical value of that allocation through substantial runtime reductions. However, lower hypervolume and worse distance indicators relative to full refinement show that some omitted calls still matter to the final front. The defensible interpretation is therefore conditional. When runtime is constrained, selective refinement offers a tunable intermediate method; when Pareto quality relative to the incumbent reference dominates computation, this experiment gives no basis for preferring H-NSGA-II over the alternatives.

The comparison also cautions against treating local-refinement success as final-front success. Hit rates exceed 98% for H-NSGA-II on I02–I04, yet its archive can remain

worse because a locally useful area change may not improve global spread, and repeated exploitation can alter diversity. Future tests should vary the refinement fraction, exploration share, and gain target, and should compare equal wall-clock budgets rather than only equal generations.

Several limits bound inference. The experiment contains four complete instances and ten stochastic seeds per method. Exact reference archives for I02–I04 contain time-limited points, so quality indicators measure agreement with an incumbent reference, not distance to a proven Pareto set. The covariance objective captures correlated price exposure but excludes weather, yield, and behavioural uncertainty. The market-throughput extension was inactive because no reproducible demand file was present. Because land utilization is neither an objective nor a constraint, part of the reduction in monetary exposure can arise through lower productive intensity, as the representative utilization audit makes explicit. Future robustness analysis should evaluate minimum space–time utilization requirements on representative instances before inferring that the reported income–risk fronts remain stable under stronger production-intensity expectations. Finally, the 17,059-row productivity source has uneven municipality–crop support; empirical eligibility reduces extrapolation but does not remove data sparsity.

9. Conclusion

This study formulates weekly agricultural production planning as a bi-objective event model with discrete crop calendars, continuous cultivated area, empirical crop eligibility, and correlated monetary price risk. A publication-safe experiment on four complete instances compares SCIP ε-constraint computation, NSGA-II, full SLSQP area refinement, and surrogate-guided selective refinement.

The evidence reveals a three-way trade-off. Exact computation provides status-bearing reference points but grows from seconds to hours and loses full-front certification as scale increases. Standard NSGA-II is the lowest-cost stochastic method and achieves the highest median hypervolume among stochastic alternatives on three instances. Surrogate gating cuts refinement calls by 75% and substantially reduces refinement and total runtime on the larger completed instances, but the savings coincide with lower median Pareto quality than full refinement. Method choice should therefore reflect the value assigned to certification, frontier quality, and response time. Larger benchmark conclusions require completion of all prescribed seeds and must be regenerated through the same snapshot rules rather than inferred from partial runs.

Data availability statement

The local reproducibility package contains source code, configuration, input provenance, benchmark metadata, result fronts, seed-level metrics, and the scripts used to regenerate every numerical table and figure. Redistribution of source data remains subject to its original provider terms.

Disclosure statement

No potential conflict of interest is reported by the author.

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